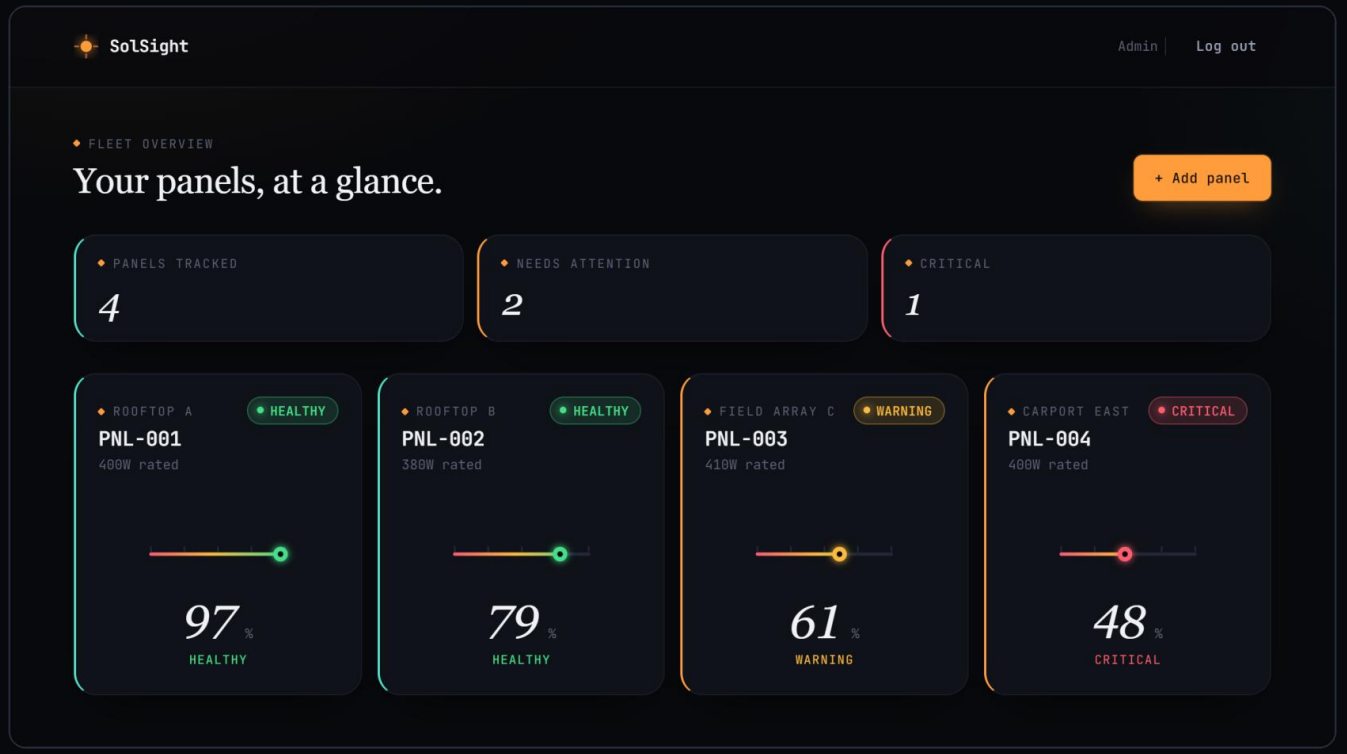
◆ PROJECT PRESENTATION

SolSight

Catching failing solar panels from their own readings

4 defect classes **90d** forecast **CNN → MLP → Bi-LSTM**



Presented by [team names] · Guide [name] · [institution]

◆ THE PROBLEM

A panel can lose a third of its output and nobody notices

01 Losses build slowly or hide inside a whole string's total, so the monthly yield barely moves.

02 Inspection is manual and periodic: someone walks the roof with a thermal camera once or twice a year.

03 By the time anyone looks, a hotspot has had weeks to cook the cells around it.

−0.4% output lost per °C of cell temperature above 25 °C, for typical crystalline silicon

1 in 3 of a panel's output gone when one cracked cell trips one of its three bypass diodes

~0.5% a year is normal ageing (NREL field-data median), so anything faster is a fault

THE IDEA

Log a reading, get a diagnosis and a forecast

01

Log a reading

Voltage, current and cell temperature from the panel, plus an optional thermal or visual photo.

02

Diagnose

The most likely defect (normal, soiling, crack or hotspot) and a health score from 0 to 100: Healthy, Warning or Critical.

03

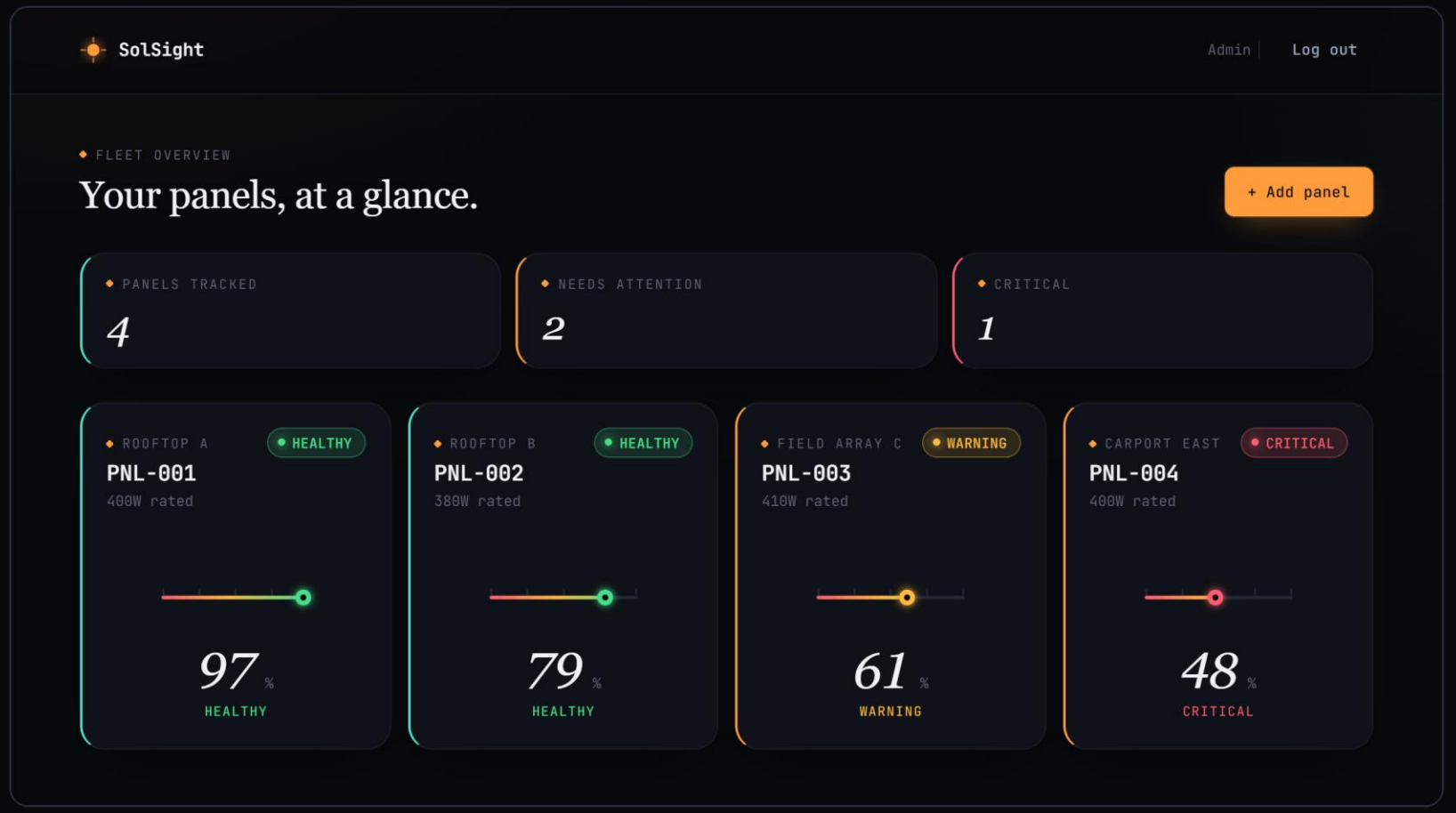
Forecast and act

Projected health at 30, 60 and 90 days, and a specific action for the technician to take.

Every result is stored against the panel, so each new reading is judged against that panel's own history rather than a fleet average.

PRODUCT

The whole fleet on one screen



TOP ROW

How many panels are tracked, how many need attention, and how many are critical.

PANEL CARDS

Each panel's latest health on a 0–100 gauge, coloured by status.

CLICK THROUGH

Opens that panel's forecast, recommendation and full inspection log.

◆ PRODUCT

Watching a hotspot develop

WEEKS 1–5

Healthy, scores 93–96. Output within a few percent of what's expected.

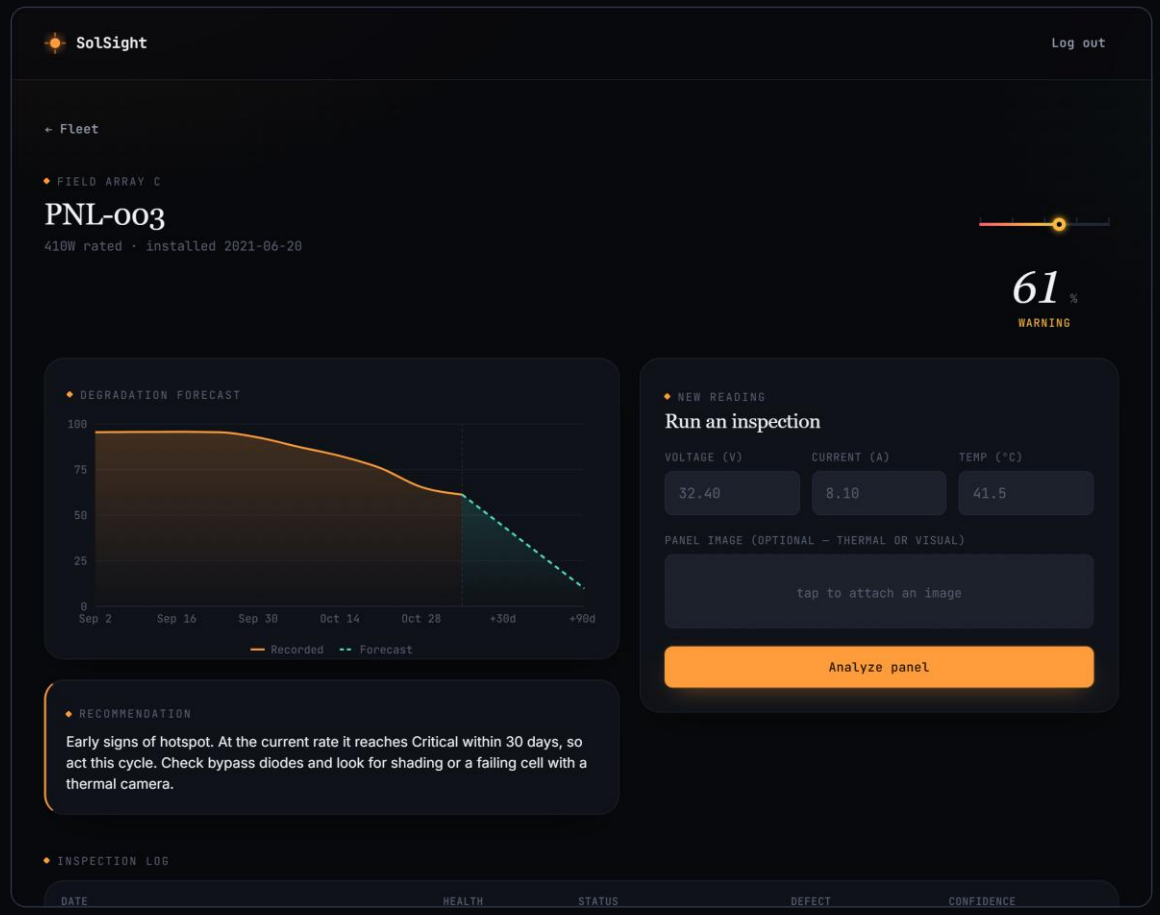
WEEKS 6–8

Output slips 10–20% while temperature stays normal, so SolSight reads it as soiling.

WEEKS 9–10

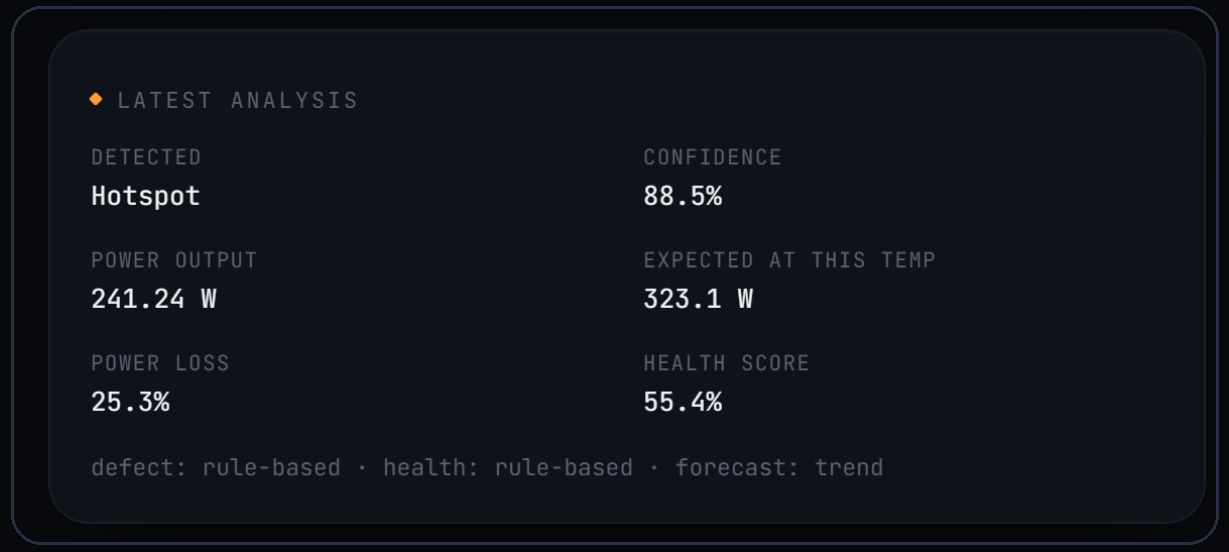
Cell temperature passes 70 °C and output keeps falling. The diagnosis switches to hotspot and the status drops to Warning.

Forecast: Critical within 30 days. Check bypass diodes and look for shading or a failing cell with a thermal camera.



PRODUCT

One reading, worked through



Result as shown in the app after submitting the reading for PNL-003

MEASURED

32.6 V × 7.4 A = 241 W

EXPECTED AT 78 °C

410 W × (1 − 0.004 × 53) = 323 W

SHORTFALL

25% below expected, while running hot

VERDICT

Hotspot (88.5%), health 55, Warning

◆ HOW IT WORKS

Three stages, each with a fallback



A stage only uses its network when trained weights are on disk. Without them it switches to the physics-based method and labels the result, so an estimate is never passed off as a model output.

◆ HOW IT WORKS

What each defect looks like in the data

DEFECT	SIGNATURE IN THE READINGS	WHAT THE TECHNICIAN IS TOLD
Normal	Output within about 10% of expected, normal cell temperature	Keep the standard schedule
Soiling	Moderate, even loss of roughly 10–25% at normal temperature	Clean the surface, then re-measure
Crack	Heavy loss of 30% or more at normal temperature, often a sudden step	Inspect for micro-cracks, plan replacement
Hotspot	Loss together with high cell temperature or a bright spot in the thermal image	Check bypass diodes and shading

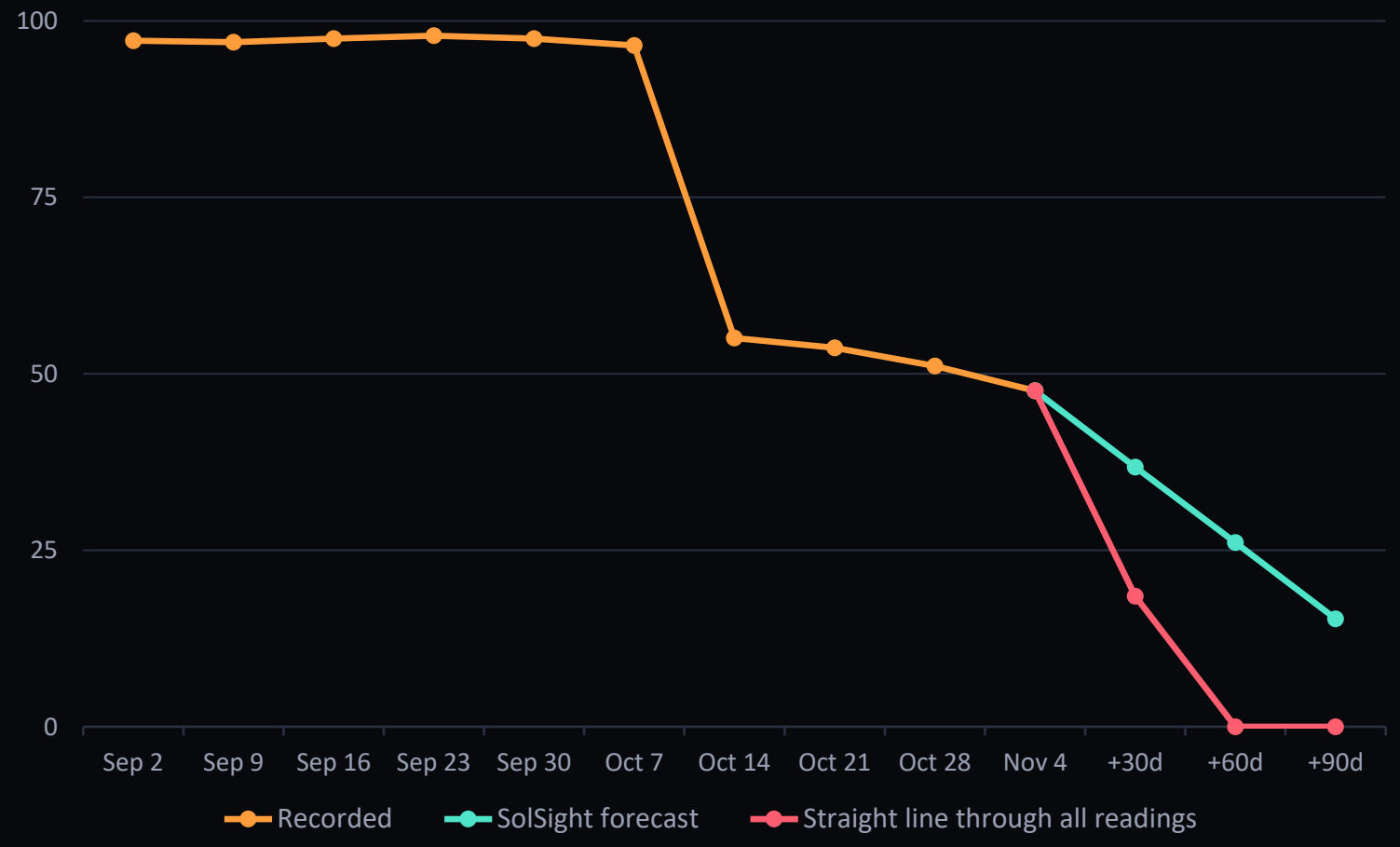
HEALTH BANDS

Healthy ≥ 75

Warning 50–75

Critical < 50

A sudden drop is an event, not a trend



In the sample data, PNL-004 cracks in week 7 and loses a third of its output between two inspections.

A straight line through all ten readings says the panel is dead within 60 days.

SolSight treats any drop of 15 points or more as an event and fits only the readings after it: 37% at 30 days, 15% at 90.

◆ HOW IT WORKS

How it's built

BROWSER

React 18 + Vite

Dashboard, panel pages and upload form. Charts with Recharts.

JSON + JWT →

API

FastAPI (Python)

Auth, panels and inspections. Models load once at startup and run off the request thread.

Motor →

DATABASE

MongoDB

Users, panels and every inspection with its full result.

MAIN ENDPOINTS

POST	/auth/login	GET	/panels	POST	/predict-defect
GET	/panels/{id}/history	GET	/panels/{id}/prediction	DELETE	/panels/{id}

Runs with Docker Compose, or with MONGO_URI=memory for a demo that needs no database at all.

◆ ENGINEERING

What we fixed while testing it

PROBLEM FOUND	FIX
Login crashed on the current bcrypt release	Password hashing rewritten on bcrypt directly
Untrained models gave random answers: a panel 70% down at 70 °C came back Healthy	Physics-based fallback, and every result says which method produced it
A saved but unanalysed reading broke the fleet page	All reads filter on analysed inspections only
A corrupt image upload caused a server error	Clear error message, and images capped at 5 MB
The forecast model was trained on a different target than it was read with	Training and inference now use the same target

22 automated tests, all passing: login, panel ownership, the full inspection flow, upload validation and the diagnostics logic.

NEXT STEPS

What comes next

01 Train the image model

Labelled crack and hotspot images, such as the public ELPV electroluminescence set, let the CNN take over stage 1 with no code change.

02 Add irradiance

We assume readings are taken near peak sun. A pyranometer value would make the expected-power baseline exact.

03 Read from the inverter

Pull voltage and current automatically on a schedule instead of typing them in.

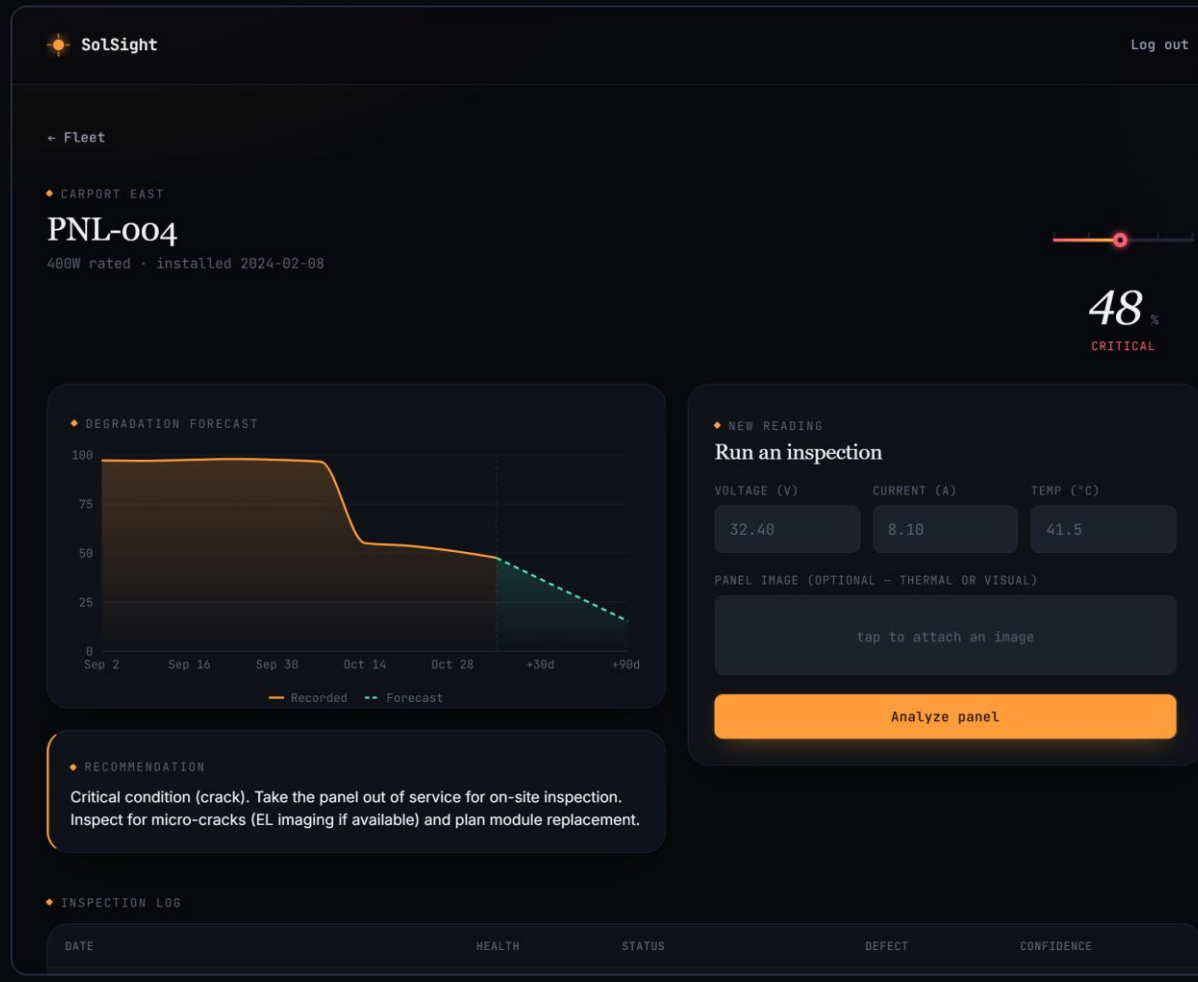
04 Alert on change

Notify the site owner when a panel crosses into Warning, or when its forecast does.

◆ THANK YOU

Questions?

github.com/SaikatxAlpha/Sol-Sight



Presented by [team names] · Guide [name] · [institution]